Notes: Petersen and Rajan (JF, 1994)

The Benefits of Lending Relationships: Evidence from Small Business Data

Contents

1 Big picture

- **Question.** How do lending relationships between small firms and their creditors affect the *price* and *availability* of credit?
- **Setting.** The paper studies U.S. small businesses using the National Survey of Small Business Finances, collected by the Small Business Administration and the Federal Reserve in 1988–1989.
- **Core friction.** Small firms are opaque. They are typically too small and young to be followed by rating agencies, analysts, or public investors, so lenders face serious adverse-selection and moral-hazard problems.
- **Main economic idea.** A bank relationship can produce private information about the firm. The relationship may reduce the lender's expected cost of lending, but the benefit may show up as more credit rather than a lower interest rate.
- **Main empirical contrast.** The paper separately studies:
  - the **cost of credit**, measured by the interest rate on the firm's most recent institutional loan;
  - the **availability of credit**, measured indirectly by costly reliance on trade credit, especially paying suppliers late or not taking early-payment discounts.
- **Main results.**
  - Relationship length has little effect on loan interest rates.
  - Purchasing financial services from a lender has little effect on loan rates.
  - Borrowing from more banks is associated with higher loan rates.
  - Stronger relationships have large effects on credit availability: firms with longer relationships, more services from lenders, and more concentrated borrowing rely less on costly trade credit.
- **Key takeaway.** Lending relationships are valuable, but they operate mainly through **quantities** rather than **prices**. Relationship lenders appear to make more credit available to opaque small firms rather than simply charging lower rates.
- **Policy implication.** If banks accumulate durable, nontransferable information about borrowers, then bank failures or forced contractions can destroy informational capital. The paper therefore connects relationship banking to credit availability and financial-system policy.

2 Data and measurement (key objects)

2.1 Observed small-business data

- **Primary data source.** The paper uses the National Survey of Small Business Finances.
- **Survey timing.** The survey was conducted in 1988 and 1989 and collected financial data for the last fiscal year.
- **Population.** The sample covers nonfinancial, nonfarm small businesses operating as of December 1987.
- **Size restriction.** Only firms with fewer than 500 employees are included.
- **Sampling design.** The survey is stratified by census region, urban/rural location, and employment size to ensure representation of rural and larger small firms.
- **Sample size.** The full sample contains **3,404 firms**: 1,875 corporations, including S-corporations, and 1,529 partnerships or sole proprietorships.
- **Typical firm.** The firms are small and young: the median book assets are about **$130,000**, median sales are about **$300,000**, and median age is about **10 years**.
- **Ownership.** Nearly 90 percent are owner-managed; about 12 percent are owned by women and 7 percent by minorities.

2.2 Firms, lenders, and borrowing sources

- **Institutional debt.** Institutional debt excludes loans from owners and families.
- **Banks.** Commercial banks, savings and loans, savings banks, and credit unions are classified as banks.
- **Nonbank financial institutions.** Finance companies, insurance companies, brokerage or mutual fund companies, leasing companies, and mortgage banks are classified as nonbank financial institutions.
- **Other borrowing sources.** The paper also observes borrowing from nonfinancial firms, venture capitalists, government agencies, owners, and family members.
- **Close lenders.** A lender is classified as "close" if the firm obtains at least one financial service from it.
- **Financial services.** These include checking and savings accounts, cash management, credit card processing, factoring, sales financing, pension-fund management, bankers' acceptances, and related services.

2.3 Relationship measures

The paper uses three main proxies for the strength of lending relationships.

- **Length of relationship.** The number of years the firm has had its longest relationship with a financial institution, or the length of the relationship with the current lender in the loan-rate regressions.
- **Multiple-service relationship.** Whether the firm buys informational financial services from its lender, or the fraction of debt borrowed from institutions that provide such services.
- **Concentration of borrowing.** The number of banks or institutions from which the firm borrows. More lenders means a less concentrated relationship.
- **Local-market concentration.** The Herfindahl index for local financial institutions, coded in three categories, proxies for how competitive the local banking market is.

2.4 Price and availability outcomes

- **Price of credit.** The interest rate on the firm's most recent institutional loan. This sample contains **1,389 firms**. Banks account for about **82 percent** of these loans.
- **Mean loan rate.** The average interest rate in the most-recent-loan sample is about **11.3 percent**, with a standard deviation of **2.2 percent**.
- **Why availability is harder.** The debt-to-assets ratio mixes demand and supply. Low debt may mean rationing, but it may also mean the firm has little need for outside finance.
- **Availability proxy.** The paper uses costly trade-credit behavior as an inverse measure of institutional-credit availability.
- **Two trade-credit variables.**
  - percentage of trade credit paid late;
  - percentage of early-payment discounts taken.
- **Interpretation.** A firm that pays suppliers late or gives up early-payment discounts is using expensive supplier financing. This suggests that ordinary institutional finance is not fully available.

2.5 Key descriptive facts

- Corporations are much larger than noncorporations: mean book assets are about **$1.7 million** for corporations versus **$0.25 million** for sole proprietorships and partnerships.

- Institutional debt-to-assets ratios are similar for corporations and noncorporations, roughly **27 percent** and **24 percent**, respectively.
- But many firms have no institutional borrowing: about **28 percent** of corporations and **45 percent** of noncorporations.
- The smallest 10 percent of borrowing firms obtain about **50 percent** of their debt from banks and about **27 percent** from owners and family.
- The largest 10 percent obtain about **62 percent** from banks and only about **10 percent** from personal sources.
- Small firms borrow from very few lenders. When the largest lender is a bank, the smallest firms borrow about **95 percent** of debt from that single bank, while the largest firms borrow about **76 percent** from that single bank.

2.6 Unobservables and timing

- **Unobserved firm quality.** Relationship variables may partly capture borrower quality, investment opportunities, or financial distress.
- **Unobserved demand for credit.** Firms using more trade credit may have better investment opportunities rather than weaker institutional-credit access.
- **Relationship timing.** Long relationships are accumulated over time, so age and relationship length are correlated. The paper includes both to separate public reputation from private relationship information.
- **Cross-sectional design.** The paper is mostly cross-sectional. It does not observe a clean experiment that randomly assigns firms to relationships.
- **Central empirical task.** The paper therefore relies on multiple relationship measures, multiple outcomes, and robustness checks to argue that relationships matter for credit availability.

3 Models and empirical specifications (all equations + interpretation)

3.1 Information theory of lending relationships

The starting point is the Stiglitz-Weiss credit-rationing logic. Raising the loan rate can worsen the borrower pool or borrower incentives. Therefore, lenders may ration credit rather than raise the price.

A lending relationship can help because it changes what the lender knows:

- over time, the lender learns whether the firm repays and whether the owner is trustworthy;
- through deposit accounts and financial services, the lender observes cash flows and transactions;
- through repeated interaction, the lender can monitor or influence the borrower more effectively.

Interpretation. Relationships can reduce adverse selection and moral hazard. But if the information is private to the relationship lender, the lender may gain monopoly power, so lower lending costs need not translate into lower interest rates.

3.2 Price regression: interest rate on the most recent loan

The paper estimates regressions of the form

$$\begin{aligned} r_i = \alpha &+ \beta_1 \text{InterestRateControls}_i + \beta_2 \text{FirmCharacteristics}_i \\ &+ \beta_3 \text{LoanCharacteristics}_i + \beta_4 \text{RelationshipCharacteristics}_i + \varepsilon_i. \end{aligned} \tag{1}$$

The dependent variable is the interest rate on the firm's most recent institutional loan.

The interest-rate controls include:

- the prime rate;
- a term-structure spread;
- a default spread, measured using BAA corporate-bond yields relative to long-term government bonds.

Firm and loan controls include:

- log book assets;
- leverage;
- corporate form;
- loan type, such as floating-rate loan;
- lender type;
- collateral type;
- region and industry dummies.

Relationship controls include:

- relationship length;
- deposit accounts and financial services with the lender;

- number of banks from which the firm borrows;
- local financial-market concentration.

Interpretation. This regression asks whether relationships lower the price of credit, conditional on observable borrower risk, loan characteristics, and macro interest-rate conditions.

3.3 Debt-ratio regression and why it is not enough

A natural first measure of credit availability is the firm's debt ratio. The paper estimates a tobit specification of the form

$$\frac{Debt_i}{Assets_i} = \alpha + X_i'\beta + R_i'\gamma + u_i, \tag{2}$$

where X_i contains firm characteristics and R_i contains relationship variables.

But this object is difficult to interpret because

$$Debt_i = \min\{\text{credit demanded by firm, credit supplied by lenders}\}. \tag{3}$$

Interpretation. A low debt ratio may mean that the firm is rationed, or it may mean that the firm simply does not want external funds. This is why the paper moves to trade-credit usage as a more informative proxy for institutional-credit constraints.

3.4 Trade-credit logic: expensive backup financing

The paper's key availability argument is summarized by Figure 1.

Let the firm have three financing sources:

- internal cash;
- institutional lending;
- an expensive alternative source, interpreted as trade credit.

If institutional finance is limited, the firm uses costly trade credit. Therefore,

$$\text{Institutional credit availability lower} \quad \Rightarrow \quad \text{costly trade-credit usage higher.} \tag{4}$$

The paper measures days payable outstanding as

$$DPO_i = 365 \times \frac{AccountsPayable_i}{CostOfGoodsSold_i}. \tag{5}$$

For example, with terms 2/10 net 30, forgoing a 2 percent discount for 20 additional days of financing implies an annualized rate of approximately

$$\left(1+\frac{2}{98}\right)^{365/20}-1\approx 44.6\%. \tag{6}$$

Interpretation. Trade credit is often far more expensive than institutional loans. Firms do not usually use it as a cheap substitute; they use it when ordinary credit is constrained or insufficient.

3.5 Availability regression: late payment of trade credit

The main availability regression is

$$\begin{aligned} LatePay_i = \alpha &+ \beta_1 \text{InvestmentOpportunities}_i + \beta_2 \text{CashFlow}_i \\ &+ \beta_3 \text{DebtCapacity}_i + \beta_4 \text{RelationshipCharacteristics}_i \\ &+ \delta_{industry} + \varepsilon_i. \end{aligned} \tag{7}$$

The dependent variable is the percentage of trade credit paid after the due date. Because the dependent variable is censored at 0 and 100 percent, the paper estimates a two-sided tobit model.

The relationship variables include:

- length of the longest relationship with a financial institution;
- fraction of borrowing from institutions that provide financial services;
- number of institutions from which the firm borrows;
- local Herfindahl index for financial institutions.

Interpretation. Lower late payment means more institutional credit availability. If relationships improve availability, relationship length and financial-service ties should reduce late payment, while borrowing from many institutions should increase late payment.

3.6 Robustness regression: early-payment discounts taken

The paper also estimates the same kind of model using the percentage of early-payment discounts taken:

$$DiscountTaken_i = \alpha + X_i'\beta + R_i'\gamma + \delta_{industry} + \varepsilon_i. \tag{8}$$

Here the expected signs are reversed relative to the late-payment regression:

$$\text{stronger relationships} \quad \Rightarrow \quad \text{more discounts taken.} \tag{9}$$

Interpretation. If relationships increase institutional-credit availability, firms should be less forced to finance themselves by stretching suppliers. They should therefore take more supplier discounts.

3.7 Robustness and alternative interpretations

The paper addresses several concerns.

- **Investment-opportunity concern.** Firms paying late may simply have better investment opportunities. The paper adds sales growth, book-assets-to-sales, industry profitability, and industry profit volatility. Relationship coefficients remain robust.
- **Industry trade-credit norms.** Some industries may tolerate late payment more than others. The paper adds late-payment stretch and the fraction of firms paying more than 50 percent late within the industry.
- **Distress concern.** Firms paying late might simply be distressed. Table X shows that firms using more trade credit do not pay substantially higher interest rates on institutional loans, which weakens this interpretation.
- **Multiple-lender quality concern.** Firms with multiple banks may be lower quality. The paper adds sales growth and interest coverage; the number-of-banks effect remains similar.

4 Identification and instruments (what makes the estimates credible)

4.1 What is hard to identify

- Lending relationships are not randomly assigned.
- Better firms may naturally have longer relationships.
- Younger firms may both have shorter relationships and more growth opportunities.
- Firms with good projects may borrow more from suppliers because they want to invest more, not because banks ration them.
- The price of loans may be rigid or administratively constrained, so relationship benefits may not appear in interest rates even when they exist.

4.2 What identifies relationship effects

The paper's identification is based on conditional correlations using detailed controls and multiple measures.

- The authors compare firms with different relationship strength after controlling for size, age, profitability, leverage, corporate form, region, industry, loan characteristics, and aggregate interest-rate conditions.
- They distinguish **firm age** from **relationship length**. Firm age captures public reputation; relationship length captures lender-specific information.
- They distinguish several dimensions of relationships: duration, breadth of services, and concentration of borrowing.
- A single story must explain why these different relationship measures all predict trade-credit behavior in the expected direction.

4.3 Why trade credit helps with identification

- Debt ratios are hard to interpret because they mix credit demand and credit supply.
- Trade-credit usage is more informative because the alternative is costly.
- Firms with adequate institutional credit should normally avoid expensive late payment or foregone discounts.
- Therefore, paying late and failing to take discounts are plausible inverse measures of institutional-credit availability.

4.4 Why the evidence is credible

- The same broad relationship story appears in both late-payment and discount-taking regressions.
- The effects are much stronger for credit availability than for interest rates, which fits theories of credit rationing and relationship-lender informational monopoly.
- Results survive controls for sales growth and industry profitability, which reduces the concern that relationship variables merely proxy for investment opportunities.
- Results survive industry trade-credit controls, which reduces the concern that some industries simply have cheaper or more tolerated supplier credit.
- Table X weakens the distress interpretation: firms that use more trade credit do not clearly pay higher institutional loan rates.

4.5 Limitations

- The paper is observational and cross-sectional, so the estimates should not be read as fully causal in the modern experimental sense.
- Relationship quality may still be correlated with unobserved borrower quality.
- The trade-credit proxy is indirect: using costly trade credit is evidence of limited institutional credit, but not a direct lender-side credit-supply measure.
- The paper cannot fully distinguish whether relationship information is private to the lender or publicly inferred from the fact that the firm has survived in a long relationship.
- The data describe small U.S. firms in the late 1980s; relationship lending may differ in periods with credit scoring, online banking, or securitized lending.

4.6 Relation to the literature

- **Theory of information and credit rationing.** The paper builds on Stiglitz and Weiss, Diamond, Fama, Sharpe, and Rajan by examining how private lender information affects credit outcomes.
- **Empirical relationship banking.** It complements studies of Japanese main banks and U.S. loan announcements by using detailed small-business relationship measures.
- **Small-firm finance.** It provides early evidence that opaque small firms benefit from concentrated relationship banking.
- **Bank competition.** It links relationship value to local financial-market concentration, foreshadowing later work on competition and relationship lending.

5 Tables and figures

5.1 Table I: sample firms by industry

Read:

- Table I describes the distribution of small firms across industries.
- Services and retail trade dominate the sample, each with roughly 930 firms.
- Manufacturing firms are fewer, about 408, but are larger on average in asset size.
- The table matters because the paper's results are about a very different population from public firms: these are small, opaque, privately held businesses.

Table I

Distribution of Sample Firms by Industry

This table contains the distribution of firms in our sample by the one-digit SIC code.

Industry	Number of Firms	Asset Size (in 1,000s of Dollars) Min.	Mean	Median	Max.	Firm Age (in Years) Mean	Median
Mining	26	30	3,129	464	32,317	12.5	7.0
Construction	447	1	708	103	12,000	14.2	12.0
Manufacturing	408	1	2,839	452	154,087	16.4	12.0
Utilities and transportation	117	7	1,778	275	62,983	13.3	10.0
Wholesale trade	344	1	1,671	302	35,945	15.1	12.0
Retail trade	930	1	589	114	22,820	12.2	9.0
Insurance and real estate	194	1	692	153	10,671	15.7	12.0
Services	938	1	591	82	69,073	13.8	10.0

Table II

Amount and Sources of Borrowing: By Size and Age

The first row is based on the smallest 10 percent of firms (book assets of less than $15,000). The asset percentiles are based on the entire sample (N = 3,404). The average debt is calculated for firms with debt only. The fraction of total borrowing, from different sources is described for firms that have debt. These percentages do not sum to 100 percent since the "not otherwise classified" category is not included. The F-statistic tests the equality of the means in each column. The last column contains the fraction of debt which firms obtain from close lenders. Close lenders are institutions which also provide the firm with at least one financial service. These include checking and savings accounts, cash management services, bankers acceptances, credit card processing, pension fund management, factoring, or sales financing.

Book Value of Assets ($1,000)	Assets Percentile	Percentage of Firms with Debt	Debt ($1,000) Mean	Debt ($1,000) Median	Fraction Borrowed from Each Source: Bank	Nonbank Financial Institution	Owner	Family	Other Firms	Fraction of Institutional Debt from Close Lenders
Panel A: Sources of Borrowing: By Size										
Less than 15	0–10	0.34	9	6	0.51	0.10	0.13	0.14	0.04	0.45
15–46	10–25	0.55	17	12	0.56	0.12	0.11	0.09	0.03	0.52
46–130	25–50	0.71	36	28	0.58	0.11	0.09	0.11	0.03	0.55
130–488	50–75	0.82	107	80	0.55	0.11	0.11	0.09	0.04	0.54
488–2,293	75–90	0.91	438	300	0.60	0.12	0.10	0.05	0.03	0.61
Over 2,293	90–100	0.91	2933	1585	0.62	0.14	0.07	0.03	0.04	0.62
F-statistic					2.10	0. 40	1.59	6.07	0.21	2.74
p-value					0.06	0.85	0.16	0.00	0.96	0.02
Panel B: Sources of Borrowing: By Age										
Less than 2	0–10	0.79	648	40	0.49	0.09	0.10	0.17	0.05	0.52
2–5	10–25	0.77	395	61	0.54	0.12	0.14	0.11	0.03	0.60
5–10	25–50	0.77	334	53	0.58	0.11	0.10	0.09	0.04	0.57
10–19	50–75	0.74	410	54	0.63	0.12	0.08	0.05	0.03	0.57
19–30	75–90	0.71	695	96	0.60	0.14	0.10	0.04	0.04	0.55
Over 30	90–100	0.59	912	128	052	0.15	0.10	0.06	0.04	0.50
F-statistic					5.72	1.19	2.24	13.10	0.97	1.24
p-value					0.00	0.31	0.05	0.00	0.44	0.29

5.2 Table II: borrowing patterns by size and age

Read:

- Small firms rely heavily on banks, owners, and family.
- As firms become larger, bank borrowing becomes more important and owner/family borrowing becomes less important.
- The fraction of institutional borrowing from close lenders rises with size, from roughly 0.45 for the smallest firms to roughly 0.62 for the largest firms.
- By age, young firms rely more on personal funds and close lenders, while older firms gradually move toward more arm's-length financing.
- This table motivates the relationship view: small firms do not borrow in anonymous capital markets; they borrow from a small set of closely connected lenders.

5.3 Table III: concentration of borrowing

Read:

- Small-firm borrowing is highly concentrated.
- When the largest lender is a bank, the smallest firms obtain about 95 percent of total debt from that single bank.
- Even the largest firms in the sample obtain about 76 percent of debt from their largest bank lender.
- Concentration declines with size, but it remains high throughout.

Table III

Concentration of Borrowing: By Size and Age

The results are reported by firm size, firm age, and primary source of debt. The asset percentiles are based on the entire sample and are the same as the ones used in Table II, Panel A. The first row contains information on the smallest 10 percent of the firms. The fraction of total borrowing from the largest single lender is reported by type of largest lender. The number of firms in each cell are reported in parentheses. The *F*-statistic tests the hypothesis that the percentage of borrowing from the largest lender is constant across the different size categories.

		Fraction of Borrowing from Largest Lender				
Book value of Asset ($1,000)	Asset Percentiles	Bank	Nonbank Financial Institution	Owner	Family	Other Firms
Panel A: Concentration of Borrowing by Size						
Less than 15	0–10	0.95 (51)	0.93 (10)	0.97 (12)	0.93 (14)	0.79 (5)
15–46	10–25	0.93 (153)	0.88 (31)	0.92 (32)	0.90 (26)	0.88 (9)
46–130	25–50	0.88 (359)	0.81 (62)	0.87 (52)	0.84 (65)	0.87 (17)
130–488	50–75	0.84 (390)	0.79 (72)	0.74 (81)	0.81 (65)	0.85 (26)
488–2,293	75–90	0.79 (296)	0.74 (49)	0.73 (38)	0.81 (23)	0.73 (15)
Over 2,293	90–100	0.76 (211)	0.72 (43)	0.75 (18)	0.74 (7)	0.71 (9)
F-statistic		19.98	2.61	6.16	1.79	1.44
p-value		0.00	0.03	0.00	0.12	0.22
Panel B: Concentration of Borrowing by Age						
Less than 2	0–10	0.86 (150)	0.80 (25)	0.85 (25)	0.89 (51)	0.76 (16)
2–5	10–25	0.85 (219)	0.77 (43)	0.82 (52)	0.83 (44)	0.86 (10)
5–10	25–50	0.85 (347)	0.76 (58)	0.81 (53)	0.83 (54)	0.84 (24)
10–19	50–75	0.85 (426)	0.80 (71)	0.80 (50)	0.78 (30)	0.88 (15)
19–30	75–90	0.82 (216)	0.84 (42)	0.73 (34)	0.77 (12)	0.70 (10)
Over 30	90–100	0.80 (106)	0.78 (28)	0.83 (19)	0.97 (9)	0.83 (6)
F-statistic		1.51	0.62	0.96	2.10	1.25
p-value		0.18	0.68	0.44	0.07	0.29

- The table motivates the later use of the number of lenders as a measure of relationship concentration.

5.4 Table IV: borrowing costs and relationship measures

Table IV

Borrowing Costs and the Role of Relationships

The dependent variable is the interest rate quoted on the firm's most recent loan. Standard errors are reported in parentheses. In addition to the variables reported, each regression also includes seven industry dummies based on the one-digit SIC codes, three regional dummies, six dummy variables for the type of assets with which the loan is collateralized, and an intercept.

Variable	(1)	(2)	(3)	(4)
Interest rate variables				
Prime rate	0.278*	0.282*	0.312*	0.278*
	(0.030)	(0.030)	(0.035)	(0.030)
Term structure spread	−0.019	−0.017	0.000	−0.027
	(0.083)	(0.083)	(0.100)	(0.083)
Default spread	0.333**	0.340**	0.183	0.325**
	(0.149)	(0.149)	(0.175)	(0.149)
Firm characteristics				
Log(book value of assets)	−0.254*	−0.264*	−0.255*	−0.259*
	(0.045)	(0.044)	(0.056)	(0.045)
Debt book assets	0.005	−0.015	−0.051	0.001
	(0.143)	(0.143)	(0.159)	(0.143)
Borrower is a corporation (0, 1)	−0.238***	−0.229***	−0.257	−0.243***
	(0.139)	(0.139)	(0.169)	(0.140)
Sales growth (1986–87)[a]			−0.585*	
			(0.301)	
Profits/interest[a]			−0.010*	
			(0.006)	
Mean 1987 gross profits/assets ratio in two-digit SIC industry				1.391**
				(0.700)
Mean σ(gross profits/assets) between 1983–87 in two-digit SIC industry				−0.771
				(0.681)
Loan characteristics				
Floating rate loan (0,1)	−0.463**	−0.448**	−0.469**	−0.473*
	(0.181)	(0.181)	(0.222)	(0.182)
Bank loan (0, 1)	0.238	0.216	0.341	0.248
	(0.225)	(0.225)	(0.270)	(0.225)
Nonfinancial firm loan firm (0, 1)	−1.125*	−1.178*	−0.513	−1.138*
	(0.360)	(0.361)	(0.430)	(0.360)

Table IV—*Continued*

Variable	(1)	(2)	(3)	(4)
Relationship Characteristics				
Length of relationship (in years)[b]	0.002	0.081	0.003	0.002
	(0.006)	(0.059)	(0.007)	(0.006)
Firm age (in years)[b]	−0.014**	−0.227*	−0.011***	−0.014**
	(0.006)	(0.078)	(0.007)	(0.006)
Information financial service (0, 1)	−0.089	−0.087	0.057	−0.087
	(0.159)	(0.158)	(0.185)	(0.159)
Noninformation financial service (0, 1)	−0.097	−0.101	−0.134	−0.104
	(0.153)	(0.153)	(0.181)	(0.153)
Deposit accounts with current lender (0, 1)	0.064	0.008	−0.041	0.061
	(0.182)	(0.186)	(0.225)	(0.182)
Number of banks from which firm borrows	0.306*	0.321*	0.303*	0.302*
	(0.085)	(0.085)	(0.096)	(0.085)
Herfindahl index for financial institutions (1, 2, or 3)	0.042	0.033	−0.024	0.033
	(0.077)	(0.077)	(0.091)	(0.077)
Number of observations	1,389	1,389	978	1,389
Adjusted R^2	0.145	0.146	0.158	0.146
Root mean squared error	2.18	2.18	2.16	2.18

[a] When profits are negative, "Profits/interest" was coded as zero. Both "Profits/interest" and "Sales growth" are truncated at their 95th percentiles (76.0 and 1.0) to limit the influence of outliers.

[b] We replace length of relationship and firm age by the natural log of one plus the length of relationship and firm age in column 2. Thus the coefficient measures the change in the interest rate due to a one percent increase in the independent variable.

*Significant at the 1 percent level.

**Significant at the 5 percent level.

***Significant at the 10 percent level.

Read:

- Table IV studies the interest rate on the most recent institutional loan.
- Larger firms pay lower interest rates. A move from the 25th to 75th percentile of assets reduces

the expected rate by about 0.59 percentage points.

- Firm age lowers rates, but relationship length with the lender does not independently reduce rates.
- Financial services from the lender also do not significantly reduce loan rates.
- The number of banks matters: borrowing from one additional bank raises the loan rate by roughly 31 basis points.
- The main message is that relationships have only limited price effects.

5.5 Figure 1 and Table V: why the paper does not use debt ratios as the main availability measure

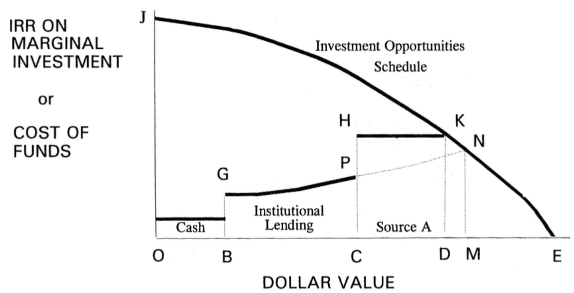

Figure 1. Sources and uses of funds. The solid curve *JKE* represents the Internal Rate of Return (*IRR*) on the marginal investment. *GP* is the marginal cost of institutional credit offered, while *HK* is the marginal cost of borrowing from *Source A*. *OB* is the amount of cash the firm has, *BC* is the amount borrowed from institutional lenders, and *CD* is the amount borrowed from *Source A*. *OD* is the total amount invested.

Table V

Determinants of the Firm's Debt Ratio

The dependent variable is total debt divided by the book value of assets. It has been multiplied by 100. Since the debt ratio is censored at zero we estimate the coefficients using a one-sided tobit model. Asymptotic standard errors are reported in parentheses. The regression also includes seven industry dummies based on the one-digit SIC codes.

Variable	
Log(book value of assets)	4.40*
	(0.56)
Profits/assets (%)	−0.99*
	(0.38)
Borrower is a corporation (0, 1)	9.32*
	(1.99)
Firm age (in years)	−0.80*
	(0.09)
Length of longest relationship (in years)	−0.19**
	(0.08)
Herfindahl index for bank deposits	3.57*
	(1.33)
Mean 1987 gross profits/assets ratio in two-digit SIC industry	19.34**
	(8.92)
Mean σ(Gross profits/assets) between 1983–87 in two-digit SIC industry	−11.94
	(9.48)
Number of observations	3,233
χ^2	347.1
(p-value)	(0.000)

* Significant at the 1 percent level.
** Significant at the 5 percent level.

Read:

- Figure 1 shows the key conceptual idea: if institutional lenders ration credit, firms turn to a more expensive alternative source.
- Table V uses the debt-to-assets ratio as the dependent variable, but the authors emphasize that this measure is ambiguous.
- Older and more profitable firms have lower debt ratios, which could mean either lower demand for debt or tighter supply.
- This motivates the move to trade credit as a better proxy for credit availability.

5.6 Tables VI and VII: trade credit as costly alternative financing

Read:

Table VI

Days Payable Outstanding and Stretch by Industry

Days Payable Outstanding = 365 ∗ Firm's Accounts Payable/Costs of Goods Sold. "Discount Stretch" is defined as the difference within the one-digit SIC industry between the median days payable outstanding for firms that take fewer than 10 percent of their early payment discounts and the median DPO for firms that take more than 90 percent of their early payment discounts. "Late Payment Stretch" is defined as the difference within the industry between the median days payable outstanding between firms that pay more than 50 percent of trade credit late and firms that pay less than 10 percent of their trade credit late. Due to the small number of observations in the utilities and transportation industry, we are not able to calculate the Discount Stretch or the Late Payment Stretch.

Industry	Number of Firms	Days Payable Outstanding: Median	Mean	Std Dev	Fraction of Firms Taking ≥ 90% of Early Payments Discounts	Discount Stretch (in Days)	Fraction of Firms Paying ≥ 50% of Trade Credit Late	Late Payment Stretch (in Days)
Mining	20	30.6	43.7	50.8	21.4	62.0	23.1	60.4
Construction	355	15.9	37.5	96.6	62.5	22.7	22.5	21.0
Manufacturing	356	27.3	43.8	83.3	42.3	14.9	24.6	18.4
Utilities and transportation	5	9.4	18.5	20.6	66.6	—	0.0	—
Wholesale trade	296	19.7	37.2	84.6	60.4	19.9	19.8	13.6
Retail trade	825	10.8	25.2	60.9	57.6	8.9	12.8	16.8
Insurance and real estate	69	0.0	207.2	1580	84.0	1.2	15.0	96.7
Services	60	4.7	16.5	26.3	56.7	16.5	12.0	0.0

Table VII

Trade Credit Usage of Firms: By Size and Age

Firms are grouped by size in Panel A and by age in Panel B. The number of firms in each cell are reported in parentheses. The *F*-statistic tests the hypothesis that the mean percentage is constant across the different size and age categories.

Book Value of Assets ($1,000)	Asset Percentiles	Percentage of Purchases Made on Credit	Percentage of Trade Credit with Early Payment Discounts	Percentage of Offered Discounts Taken by Firm	Trade Credit Paid Late in Percent	Median Stretch for Firms (Measured from Last Day for Discounts)[a]	Median Stretch for Firms (Measured from Due Date)[b]
Panel A: Trade Credit Usage of Firms: By Size							
Less than 15	0–10	71.8 (203)	33.2 (204)	74.2 (122)	18.0 (100)	−13.03 (108)	−17.91 (102)
15–46	10–25	73.3 (366)	31.4 (364)	64.4 (245)	21.2 (194)	−7.32 (245)	−10.71 (239)
46–130	25–50	74.0 (639)	32.7 (632)	63.8 (457)	19.8 (344)	−2.60 (445)	−5.86 (439)
130–488	50–75	81.2 (694)	32.8 (686)	64.9 (539)	21.7 (379)	1.44 (473)	−0.81 (464)
488–2293	75–90	84.8 (446)	31.5 (433)	68.3 (375)	21.4 (260)	6.59 (305)	3.75 (308)
Over 2293	90–100	90.2 (298)	34.1 (289)	70.5 (278)	21.0 (172)	5.27 (214)	2.85 (210)
F-statistic		25.65	0.27	2.17	0.56		
p-value		0.00	0.93	0.05	0.73		
Panel B: Trade Credit Usage of Firms: By Age							
Firms Age (Years)	Age Percentile						
Less than 2	0–10	75.4 (299)	33.6 (295)	58.9 (224)	25.1 (155)	−2.20 (208)	−5.86 (200)
2–5	10–25	74.3 (414)	32.2 (400)	57.1 (295)	23.7 (243)	0.00 (282)	−3.16 (276)
5–10	25–50	79.2 (623)	30.4 (615)	61.5 (457)	22.1 (357)	−0.02 (400)	−2.97 (398)
10–19	50–75	80.6 (689)	33.3 (682)	68.9 (539)	18.4 (392)	0.00 (470)	−2.71 (473)
19–30	75–90	82.7 (377)	33.4 (376)	74.5 (308)	18.6 (210)	2.89 (260)	−1.22 (252)
Over 30	90–100	83.5 (244)	34.1 (240)	82.4 (193)	15.8 (92)	0.11 (170)	−0.72 (163)
F-statistic		6.47	0.69	14.89	4.09		
p-value		0.00	0.63	0.00	0.00		

[a] For each two-digit SIC industry, the median DPO is obtained for firms availing of more than 90 percent of their discounts. This is subtracted from the DPO for the firm to obtain the stretch as measured from the last day for discounts.

[b] For each two-digit SIC industry, the median DPO is obtained for firms paying less than 10 percent of credit late. This is subtracted from the DPO for the firm to obtain the stretch as measured from the due date.

- Table VI documents days payable outstanding and industry-level trade-credit stretches.
- The 2/10 net 30 example implies a very high implicit annualized cost of trade credit, about 44.6 percent before considering additional late-payment costs.
- Table VII shows that most firms make purchases on trade credit.
- Late payments decline with firm age: the youngest firms pay about 25.1 percent of trade credit late, while the oldest firms pay about 15.8 percent late.
- Discounts taken rise with age: the youngest firms take about 58.9 percent of offered early-payment discounts, while the oldest take about 82.4 percent.
- These patterns are consistent with younger firms facing more institutional-credit constraints.

5.7 Table VIII: credit availability and relationships using late payments

Read:

- Table VIII is the paper's central availability table.
- A longer lending relationship reduces late payment. In the baseline late-payment regression, the coefficient on relationship length is about −0.155.
- Borrowing from institutions that provide financial services reduces late payment sharply: the baseline coefficient is about −5.576.
- Borrowing from more institutions increases late payment: one additional institution raises late payments by about 1.9 percentage points.
- Local banking concentration reduces late payment: firms in more concentrated financial markets are less likely to be rationed.
- The signs are exactly what the relationship-lending view predicts.

Table VIII

Credit Availability and the Role of Relationships
Dependent Variable: Trade Credits Repaid Late (%)

The dependent variable is the percentage of trade credits that were paid after the due date (paid late). The coefficient estimates are from a tobit regression with two-sided censoring. The dependent variable is censored at 0.0 and 1.0. Asymptotic standard errors are in parentheses. Each regression also includes seven industry dummies based on the one-digit SIC codes.

Independent Variable	(1)	(2)	(3)	(4)	(5)	(6)
Log(book value of assets)	−0.662	−0.542	−0.753	−0.678	−0.533	−1.580*
	(0.490)	(0.495)	(0.486)	(0.513)	(0.489)	(0.566)
Profits/book assets	−0.679	−0.785	−0.663	−0.643	−0.744	0.372
	(0.753)	(0.754)	(0.753)	(0.782)	(0.750)	(0.916)
Debt from institutions/ book assets	4.853**	4.902**	4.888**	4.463**	5.775*	5.932*
	(2.040)	(2.038)	(2.045)	(2.108)	(2.045)	(2.310)
Firm is a corporation (0, 1)	2.819***	2.857***	2.827***	3.248***	2.597	5.267*
	(1.635)	(1.633)	(1.638)	(1.726)	(1.632)	(1.936)
Firm age (in years)[a]	−0.142***	−0.150***	−1.531	−0.181**	−0.140***	−0.164***
	(0.081)	(0.081)	(1.080)	(0.084)	(0.081)	(0.093)
Length of longest relationship (in years)[a]	−0.155**	−0.152**	−2.299**	−0.165**	−0.150**	−0.177**
	(0.069)	(0.069)	(1.089)	(0.070)	(0.068)	(0.078)
Debt from financial service provider (%)	−5.576*		−5.385*	−5.555*	−5.713*	−6.976*
	(1.672)		(1.678)	(1.748)	(1.663)	(1.907)
Debt from financial service provider (only one service) (%)		−7.630*				
		(2.085)				
Debt from financial service provider (multiple services) (%)		−3.895**				
		(1.958)				
Number of institutions from which firm borrows	1.926**	1.872**	1.991**	2.286**	1.840**	1.546
	(0.900)	(0.900)	(0.901)	(0.946)	(0.897)	(1.043)
Herfindahl index for financial institutions (1, 2, or 3)	−2.253*	−2.300*	−2.274*	−2.065**	−1.901**	−2.659*
	(0.866)	(0.865)	(0.866)	(0.902)	(0.866)	(0.996)
Sales growth (1986–1987)				−1.445		
				(1.086)		
Mean 1987 gross profits/assets ratio in two-digit SIC industry					−28.245*	
					(8.643)	
Mean σ(gross profits/assets) between 1983–87 in two-digit SIC industry					22.677*	
					(7.851)	
Late payment stretch (in days)[b]						0.012
						(0.019)
Fraction of firms paying $\geq$ 50% of trade credit late						84.997*
						(5.758)
Number of observations	1,119	1,119	1,119	1,019	1,119	857
χ^2	83.5	86.2	80.7	86.3	96.6	127.1
(p-value)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

[a] We replace length of relationship and firm age by the natural log of one plus the length of relationship and firm age in column 3. Thus the coefficient measures the change in the interest rate due to a one percent increase in the firm's age or the length of its longest relationship.

[b] For each two-digit SIC industry, the median DPO is obtained for firms paying less than 10 percent of credit late. This is subtracted from the DPO for firms paying more than 50 percent of credit late to obtain the late payment stretch.

* Significant at the 1 percent level.

** Significant at the 5 percent level.

*** Significant at the 10 percent level.

- The magnitude is stronger than in the price regressions. Relationship length has about 138 percent of the impact of firm size on availability, while it has essentially no impact on loan price.

5.8 Table IX: credit availability using discounts taken

Table IX

Credit Availability and the Role of Relationships
Dependent Variable: Offered Discounts Taken by the Firm (%)

The dependent variable is the percentage of early payment discounts that are taken by the firm. The coefficient estimates are from a tobit regression with two-sided censoring. The dependent variable is censored at 0.0 and 1.0. Asymptotic standard errors are in parenthesis. Except for column 6, each regression also includes seven industry dummies based on the one-digit SIC codes.

Independent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log(book value of assets)	6.795*	6.554*	7.650*	6.747*	6.541*	8.549*	6.750*
	(1.679)	(1.691)	(1.670)	(1.778)	(1.678)	(2.718)	(1.751)
Profits/book assets	7.485*	7.657*	7.706*	6.493**	7.593*	14.022**	8.144*
	(2.399)	(2.419)	(2.428)	(2.624)	(2.373)	(5.601)	(2.600)
Debt from institutions/ book assets	−14.178**	−14.326**	−14.825**	−16.020**	−15.736**	−34.934**	−14.326**
	(7.161)	(7.168)	(7.217)	(7.437)	(7.184)	(13.736)	(7.353)
Firm is a corporation (= 1 if yes)	−9.913***	−9.844***	−9.811***	−12.751**	−8.831	−29.610*	−8.643
	(5.697)	(5.696)	(5.732)	(6.085)	(5.713)	(11.277)	(5.968)
Firm age (in years)[a]	0.900*	0.911*	6.612***	0.972*	0.899*	1.085*	1.003*
	(0.246)	(0.246)	(3.474)	(0.271)	(0.246)	(0.392)	(0.254)
Length of longest relationship (in years)[a]	0.904*	0.901*	17.154*	0.878*	0.884*	0.598***	0.841*
	(0.219)	(0.219)	(3.689)	(0.227)	(0.218)	(0.320)	(0.224)
Debt from financial service provider (%)	5.655		4.415	5.875	5.919	10.517	5.650
	(5.667)		(5.705)	(5.969)	(5.660)	(9.670)	(5.817)
Debt from financial service provider (only one service) (%)		10.159					
		(6.939)					
Debt from financial service provider (multiple services) (%)		1.539					
		(6.731)					
Number of institutions from which firm borrows	−8.889*	−8.790**	−9.754*	−9.486*	−8.789*	−11.192**	−7.505**
	(3.152)	(3.152)	(3.165)	(3.331)	(3.155)	(5.194)	(3.272)
Herfindahl index for financial institutions (1, 2, or 3)	14.996*	15.111*	14.608*	15.658*	14.170*	16.202*	14.449*
	(3.033)	(3.035)	(3.050)	(3.189)	(3.038)	(5.140)	(3.136)
Sales growth (1986–1987)				1.254			
				(3.744)			
Mean 1987 gross profits/assets ratio in two-digit SIC industry					78.292**		
					(33.676)		
Mean σ(gross profits/assets) between 1983–87 in two-digit SIC industry					−69.090**		
					(27.350)		
Interest rate implied by trade credit terms in two-digit SIC industry						−0.017	
						(0.063)	
Discount stretch (in days)[b]							−0.036
							(0.190)
Fraction of firms taking ≥ 90% of early payments discounts							146.60*
							(29.188)
Number of observations	1500	1500	1500	1362	1500	545	1328
χ^2	194.1	195.3	185.7	183.1	202.8	76.7	216.5
(p-value)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

[a] We replace length of relationship and firm age by the natural log of one plus the length of relationship and firm age in column 3. Thus the coefficient measures the change in the interest rate due to a one percent increase in the firm's age or the length of its longest relationship.
[b] For each two-digit SIC industry, the median DPO is obtained for firms taking more than 90 percent of discounts offered. This is subtracted from the DPO for firms taking less than 10 percent of discounts offered to obtain the discount stretch.
* Significant at the 1 percent level.
** Significant at the 5 percent level.
*** Significant at the 10 percent level.

Read:

- Table IX repeats the availability analysis using the fraction of early-payment discounts taken.
- The expected signs reverse relative to Table VIII: stronger relationships should increase discounts taken rather than reduce them.
- The results confirm the late-payment evidence.
- This robustness check is important because it uses a second measure of costly trade-credit dependence.

5.9 Table X: are trade-credit users simply distressed?

Read:

Table X

Average Interest Rates on Most Recent Loan

The table contains the average interest rate on the firm's most recent loan categorized by the firm's book value of assets, the percent of trade credits paid late, and the percentage of early payments taken. The first row of the table contains the smallest firms, the firms that pay the largest percent of their trade credits late, and the firms that take advantage of the fewest early payment discounts.

	Percentiles Based on the Book Value of Assets	Percentiles Based on the Percent of Trade Credits Paid Late	Percentiles Based on the Percent of Early Payment Discounts Taken
0–10%	12.0	11.4	11.2
10–25%	11.5	11.9	11.1
25–50%	11.5	11.4	11.1
50–75%	10.8	11.2	10.8[a]
75–90%	10.5	10.7	10.8
90–100%	10.1	10.4	10.8

[a] Over 25 percent of the firms take all of the early payment discounts that are offered. Thus the groups 50–70 percent, 75–90 percent, and 90–100 percent are not distinct. Thus 10.8 percent is the average interest rate for firms taking more than the median percent of the early discounts which they are offered.

- Table X reports average interest rates by firm size, late-payment intensity, and discount-taking behavior.
- Firms using the most trade credit do not clearly pay higher institutional loan rates.
- For example, average rates by late-payment percentiles are roughly 11.4 percent at the lowest late-payment percentile and 10.4 percent at the highest percentile.
- This weakens the interpretation that trade-credit usage is simply distress showing up everywhere in credit markets.
- Instead, it supports the interpretation that trade-credit usage captures limited institutional-credit availability.

6 Short summary

- **Method.** The paper uses small-business survey data to relate lending relationships to loan prices and credit availability.
- **Measurement.** Relationship strength is measured by relationship length, financial services purchased from lenders, and borrowing concentration.
- **Price result.** Relationships have weak effects on interest rates. The strongest price result is that borrowing from more banks raises the loan rate.
- **Availability result.** Relationships strongly improve credit availability. Firms with stronger, more concentrated relationships rely less on costly trade credit.
- **Economic result.** The value of relationship banking is mainly that it relaxes quantity constraints, not that it lowers quoted loan rates.

7 Conclusion

7.1 Core conclusion

- Lending relationships matter for small firms.
- The main benefit is greater access to credit, not necessarily cheaper credit.
- Relationships appear to help lenders produce and use information about opaque borrowers.
- Concentrated borrowing and financial-service ties are not necessarily signs of lender exploitation; they may be mechanisms through which firms obtain credit.

7.2 Economic intuition

- Small firms are informationally opaque, so arm's-length lenders may ration them.
- A relationship lender learns over time and through multiple services.
- This information can support additional lending.
- But if the information is private, the lender may not fully pass cost savings through in lower interest rates.
- Thus, the borrower gains mainly by getting more credit, not by paying a visibly lower rate.

7.3 Takeaway

Petersen and Rajan's paper is a foundational empirical study of relationship banking. It shows that relationship lending is especially important for opaque small firms and that the correct empirical margin is often credit availability rather than loan price. The paper's central lesson is simple but powerful: when information is private and lending markets are imperfect, the value of a banking relationship is not just cheaper money; it is access to money at all.